www.csvtuonline.com

Printed Pages– 7

Roll No. 311241305 7

324655(24)

B. E. (Sixth Semester) Examination, Nov.-Dec. 2016

(New Scheme)

(Elect. Engg. Branch)

PRINCIPLES of DIGITAL SIGNAL PROCESSING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *All questions are compulsory. Part (a) of each question is compulsory. Attempt any two parts from b, c and d of each question.*

Unit - I

1. (a) The accumulator $y(n) = \sum_{k=-\infty}^{n} x(k)$ is excited by

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

the sequence $x(n) = n\,u(n)$. Determine its output under the condition that it is initially relaxed. Comment upon the stability of the system. 2

(b) What are the major classification of signals? Write their mathematical expressions. Find the even and odd components of the following signals : 7

(i) $x(t) = u(t)$

(ii) $x(t) = \sin\left(\omega_0 t + \frac{\pi}{4}\right)$

(ii) $x(n) = \delta(n)$

(c) Explain the modelling a system with : 7

(i) Differential/difference equation

(ii) Impulse response

(iii) State variables

(d) What is meant by sampling? State the sampling theorem. Explain how sampling can be done with an impulse function. Draw the spectrum of the sampled signal and explain aliasing. Explain quantization. 7

www.csvtuonline.com

www.csvtuonline.com

Unit - II

2. (a) For a LTI system derive the expression for convolution sum. 2

(b) Find the frequency response of the linear shift invariant system with the unit sample response $h(n)$ as : 7

(i) $h(n) = \delta(n) + 6\delta(n-1) + 3\delta(n-2)$

(ii) $h(n) = \left(\frac{1}{3}\right)^{n+2} u(n-2)$

(c) Determine the range of values of the parameter a for which the linear time invariant system with impulse response 7

$h(n) = a^n u(n)$ is stable.

(d) If the input to the linear shift invariant system is :

$$x(n) = \left(\frac{1}{2}\right)^n u(n) + 2^n u(-n-1)$$

the output is

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

$$y(n) = 6\left(\frac{1}{2}\right)^n u(n) - 6\left(\frac{3}{4}\right)^n u(n)$$

Find the system function $H(z)$ and determine whether or not the system is stable and/or causal. 7

Unit - III

3. (a) What is discrete time Fourier transform and how it is related to Z-transform? 2

(b) Consider the sequence 7

$$x(n) = \delta(n) + 2\delta(n-2) + \delta(n-3)$$

(i) Find the four point DFT of $x(n)$

(ii) If $y(n)$ is the four point circular convolution of $x(n)$ with itself, find $y(n)$ and the four-point DFT $Y(k)$.

(c) What is the need of FFT algorithm? Given

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

$$X(k) = \{36, -4 + j9\cdot656, -4 + j4,$$

$$-4 - 4 + j1\cdot656, -4 - j1\cdot656, -4 - j9\cdot656\}$$ 7

Find $x(n)$ with 8 points inverse FFT, using DIF algorithm.

(d) Perform the linear convolution of the following sequences using :

(i) Overlap-add method and

(ii) Overlap save method.

$x(n) = \{1, -1, 2, 1, 2, -1, 1, 3, 1\}$ and

$h(n) = \{1, 2, 1\}$

Also verify your result with direct convolution. 7

Unit - IV

4. (a) What are the requirements for an analog filter and digital filter to be stable and causal? 7

(b) What is bilinear transformation? Obtain the mapping from s-plane to z-plane. Derive the relation between analog and digital freuency in bilinear transformation. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) Design a Chebyshev IIR digital low-pass filter to satisfy the constraints :

$$0\cdot707 < \left| H\left(e^{j\omega}\right) \right| < 1; \quad 0 \le \omega \le 0\cdot2\pi$$

$$\left| H\left(e^{j\omega}\right) \right| < 0\cdot1; \quad 0\cdot5 \le \omega \le \pi$$

using bilinear transformation and assuming $T = 1$ sec. 7

(d) Consider the causal linear shift invariant system with system function

$$H(z) = \frac{1 + 0\cdot875\, z^{-1}}{\left(1 + 0\cdot2\, z^{-1} + 0\cdot9\, z^{-2}\right)\left(1 - 0\cdot7\, z^{-1}\right)}$$ 7

Draw the signal flow graph for the system using

(i) Direct form I and II

(ii) A cascade of first and second order system realized in direct form II.

(iii) A parallel connection of first and second order system realized in direct form II.

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Unit - V

5. (a) What are FIR filters. Illustrate two simplest structures for implementing an FIR system. 2

(b) Given a three stage lattice filter with coefficients $K_1 = \frac{1}{4}, K_2 = \frac{1}{2}, K_3 = \frac{1}{3}$, determine the FIR filter coefficients for the direct form structure. 7

(c) List the advantages and disadvantages of FIR filters. Write in steps the procedure for designing FIR filters using windows. 7

(d) Explain in steps the technique for designing a digital filter directly using Pade's approximation. 7

www.csvtuonline.com

www.csvtuonline.com